# Issac Roy

U.S. Citizen, San Diego, CA

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## Education

**University of California San Diego (3.97 GPA)** **March 2027**

***B.S. Data Science*** | *Math Minor* | *Cognitive Science Minor* *San Diego, CA*

## Experience

**MC Computer Vision Lab at UCSD** | *Research Assistant* **January 2026 – Present**

- Building an OpenRooms++ generative pipeline on NRP's Kubernetes cluster, chaining Trellis2 and Flux 2.0 to reconstruct photorealistic 3D assets from single-view input images.
- Implementing a coarse-to-fine refinement workflow that upgrades initial meshes into detailed PBR textures (Albedo, Roughness) by enforcing multi-view consistency with SAM3D and Laval Indoor HDR.

**Neurocrine Biosciences** | *Data Engineer Intern* **June 2025 – September 2025**

- Saved $200k in annual OpEx by architecting, building, and deploying an end-to-end serverless containerized data ingestion pipeline on AWS ECS and Lambda, eliminating dependency on third-party vendors.
- Saved 50+ hours per month by developing an AI-driven data pipeline with AWS Bedrock, integrating LLM endpoints with AWS Redshift to convert raw patient data into study-specific narratives, streamlining reporting for study teams.
- Implemented DevOps workflows using Terraform (IaC), AWS Lambda, and multi-environment CI/CD pipelines, collaborating with Jira and Confluence to support Clinical Solution's mission to enhance monitoring of clinical trials.

**San Diego County Taxpayers Association (SDCTA)** | *Data Science Intern* **October 2024 – January 2025**

- Built an NLP-driven policy analysis pipeline, utilizing Playwright for data scraping and Hugging Face Transformers for sentiment analysis, supporting evidence-based recommendations for policymakers.
- Designed interactive visualizations using Python (Pandas, Plotly Dash) to communicate complex fiscal data to non-technical stakeholders, increasing transparency on municipal spending trends.

## Leadership

**Data Science Student Society (DS3)** | *Director of Software* **October 2024 – Present**

- Centralized data workflows and analytics for 1,000+ members by implementing robust Supabase authentication and architecting a normalized Postgres database with RLS policies.
- Directed a 10 engineer team to deliver a seamless UI/UX, building a large React codebase and automating deployments through custom GitHub Actions CI/CD pipelines.

**AWS Cloud Club at UCSD** | *Founder & VP Technical* **November 2025 – Present**

- Established UCSD's official AWS affiliate organization; designing and leading technical workshops on MLOps (Git, AWS, Docker, Terraform) to equip 200+ students with the practical skills to build and deploy scalable data and ML solutions.

## Projects

**Neural Nector** | *PyTorch, React, TypeScript, Python, FastAPI, Postgres* **December 2025**

- Trained a Generative Adversarial Network (GAN) to synthesize realistic flower images, achieving a FID score of 50.
- Built an interactive, full-stack web app challenging users to distinguish between real and GAN-generated flower images, engaging 100+ unique users.

**OnlyDance** | *OpenCV, MediaPipe, React Three Fiber, TypeScript, Python, FastAPI* **April 2025**

- Won 1st place at DataHacks 2025 against 50+ teams by developing an interactive full-stack app that teaches dance routines using 3D avatars and real-time feedback.
- Engineered a real-time similarity scoring algorithm using MediaPipe to extract 3D pose landmarks, implementing sliding window techniques and frame interpolation to quantify dance accuracy.

**Predicting Power Outage Severities** | *Python, Scikit-Learn, Pandas, Folium* **December 2024**

- Created an ML pipeline to predict power outage severities across the U.S. with 78% accuracy, performing data cleaning, EDA, hypothesis testing, feature engineering, and fairness evaluations across states.

**Self Playing Guitar** | *Arduino, CAD, Laser Cutting, 3D Printing* **December 2023**

- Won 1st place at IEEE's Fall Competition against 30+ teams by engineering a self-playing guitar with custom laser-cut and 3D printed parts, programming a note transcriber algorithm to map musical inputs to precise servo motions.

## Skills

**Languages:** Python, SQL, R, React, TypeScript, Java, C++
**Certifications:** AWS Cloud Practitioner, AWS AI Practitioner, AWS Solutions Architect Associate
**Tools:** AWS, Git, MongoDB, Jira, Confluence, Terraform, PyTorch, SKLearn, Docker, Linux, Spark, Selenium, PowerBI
**Hobbies:** 3D Printing, Robotics, Guitar, Rubik's Cube (PB: 26.50s), Pool (Tracking every shot for 700+ games)